Arunakiri Venkatachalam

Data Scientist | Full Stack Developer | arunakiri.github.io

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Professional Summary

Proactive and dedicated Data Scientist with over 7 years of experience in full-stack development, specializing in building scalable data pipelines, cloud-native applications, and advanced analytics solutions. Experience in orchestrating complex environments and engineering high-frequency ETL pipelines.

Experience

DATA SCIENTIST **VOLVO | OCTOBER 2023 – PRESENT**

- **Data Science Lab (DSL) Operations:** Operate within a secure, cloud-based environment using Kubernetes, Docker, and Apache Airflow for orchestration.
- **Data Pipeline Engineering:** Built high-frequency Apache Airflow ETL pipelines (running every 5 mins) for Hydrogen Engine Rigs and Test Fleet data, standardizing proprietary formats.
- **Application Development:** Developed Streamlit and Plotly Dash applications hosted on Kubernetes for 200+ users, enabling powertrain driveline analysis and hydrogen engine outlier detection.
- **Admin Solutions:** Created a full-stack Django-based user management portal with LDAP integration and multi-step approval workflows for DSL resource allocation.

LEAD ANALYST (CONSULTANT AT VOLVO) **HCL | OCTOBER 2021 – OCTOBER 2023**

- **Modernization & Cost Reduction:** Collaborated in MATLAB-to-Python modernization global initiative, reducing MATLAB dependency and licensing costs by 30%.
- **Solution Conversion:** Successfully converted complex MATLAB solutions (Dolly Tire Performance, GPS Track Visualization, Battery SOC Analysis) into efficient Python equivalents.
- **Asset Management:** Contributed to IT asset management system for tracking allocation and utilization.

SENIOR SYSTEM ENGINEER **INFOSYS | APRIL 2018 – OCTOBER 2021**

- **OCR & Data Extraction:** Developed image-scanning modules for Novartis Pharma using Tesseract, achieving 80% extraction accuracy from diverse medical leaflets.
- **Data Visualization:** Designed a PostgreSQL-based digital repository and developed Power BI "PIL Boards" for structured analysis of medical product information.

Education

Bachelors in **Electronics and Communications** | CEG, Anna University, Chennai 2016
Masters in **Data Science And Engineering** | BITS Pilani, Hyderabad 2022

Technical Skills

- **Language & Frameworks:** Python, Django Web Framework, Shell, REST API, SQL Alchemy
- **DevOps & Cloud:** Docker, Kubernetes, Git/GitHub, CI/CD DevOps, OAuth 2.0
- **Visualization & Analytics:** Power BI, Grafana, Plotly Dash, Streamlit
- **Data Science:** Machine Learning, Artificial Intelligence, Image/Text Scanning (Tesseract)